COS40007 Artificial Intelligence Engineering

Portfolio Assessment 2: Advanced Data Preprocessing, Feature Engineering & Model Development

Due: 01/04/2026 23:59 PM on Canvas

AIM

This task aims to demonstrate your familiarity with Python programming, your favourite IDE, engineering datasets and programming machine learning with Python. The task is to understand data exploration, data labelling, feature engineering, basic machine learning model development with DR and Hyperparameter tuning.

Task 1: Target Variable Preparation

1.1 For Numerical Target Variables:

Define 3 to 5 meaningful class labels by discretising your numerical target variable. Ensure the class distribution is approximately balanced when defining your bins/thresholds.

OR

1.2 For Categorical Target Variables:

If your target is already categorical, verify class distribution. If a severe imbalance (>80% in one class) exists, document this challenge and note how it will affect modelling.

Task 2: Feature Normalisation & Transformation

2.1 Normalise Numerical Features:

Apply appropriate scaling to all numerical features based on their distribution.

Distribution Type	Recommended Scaling	Code Example
Normal/Gaussian	StandardScaler	`StandardScaler().fit_transform(X)`
Skewed/non-Gaussian	MinMaxScaler or RobustScaler	`MinMaxScaler().fit_transform(X)`
Presence of outliers	RobustScaler	`RobustScaler().fit_transform(X)`

2.2 Encode Categorical/Other Features:

Convert all non-numerical features to appropriate numerical representations.

Feature Type	Encoding Method	When to Use
Nominal (no order)	One-Hot Encoding	Few categories (<10)
Nominal (many categories)	Label Encoding or Frequency Encoding	Many categories (>10)
Ordinal (has order)	Ordinal Encoding	Education level, ratings
Binary	Label Encoding (0/1)	Yes/No, True/False

2.3 Clearly document:

- Which features were scaled

- Which features were encoded
- Why were those transformations selected

Briefly explain how scaling may impact model performance.

Task 3: Feature Engineering

Using insights from your EDA:

- Create new engineered features, such as:
  - Ratios
  - Interaction terms
  - Aggregated metrics
  - Domain-specific composite indicators.
- Justify each new feature:
  - What pattern from EDA motivated it?
  - What engineering meaning does it have?
- Top Correlations, Multicollinearity and Insights can be written here.

Feature Engineering Ideas by Domain:

Engineering Domain	**Feature Engineering Ideas**
Mechanical/Civil	Ratios (strength/weight), interaction terms, polynomial features, and log transformations of exponential relationships
Electrical	Power calculations, frequency domain features, signal averages, peak-to-peak values
Chemical/Process	Reaction yields, efficiency metrics, composite indices, rate calculations
Biomedical	BMI from height/weight, risk scores, normalised vital signs, change over time
Industrial/Manufacturing	Downtime ratios, maintenance intervals, efficiency metrics, utilisation rates

Task 4: Decision Tree Modelling & Comparison

Create and evaluate Decision Tree models using at least five different feature sets derived from your EDA.

Feature Set Ideas:

Set	Description	Example
Set 1	Original features only	All original predictors
Set 2	Top correlated features	Top 5 features from correlation with the target
Set 3	Original + engineered features	Original + new features from Task 3
Set 4	PCA/ dimensionality reduced features	First 5 principal components
Set 5	Domain-selected features	Features chosen based on engineering knowledge
Set 6	Minimal viable set	Smallest set that maintains performance

Task 5: Results Comparison Table

Follow Studio 3, create a comprehensive comparison table of your five Decision Tree models.

Task 6: Model Improvement

Follow Studio 3, use SVM to improve the model and implement 5-fold cross-validation.

Task 7: Hyperparameter Tuning

Follow Studio 3, implement on your dataset.

Task 8: Feature Selection

Follow Studio 3, implement on your dataset.

Task 9: Dimensionality Reduction

Follow Studio 3, implement on your dataset.

Task 10: Final Results Table

Follow Studio 3, implement on your dataset.

Generative AI Use Declaration

As AI engineering professionals, transparent and ethical use of AI tools is a core competency. For this portfolio assessment, you may use Generative AI (e.g., ChatGPT, GitHub Copilot, Claude) for:

Permitted Uses:

- Generating code comments and documentation
- Debugging assistance
- Explaining error messages
- Suggesting features of engineering ideas
- Improving code readability

Required Declaration:

Include the following statement at the beginning of your portfolio submission:

"I used [GenAI tool name] to assist with [specific tasks, e.g., commenting code, debugging, suggesting feature combinations]. All final code, interpretations, and conclusions are my own work after verification."

Failure to declare GenAI use will result in a 10-point deduction.

Submission Requirements

Format: Single Jupyter notebook/Python file and PDF report containing:

- All code cells with outputs
- First Comparison Table with visuals
- Markdown cells explaining your reasoning
- Final Comparison Table with visuals
- GenAI Declaration at the beginning/last

File Naming:

1. Portfolio_Assessment_2_[Studio_Number]_[YourName].pdf
2. GitHub Classroom link to push your code: https://classroom.github.com/a/ifqEa70j